

CLOUDEAR: INTEGRATING PERCEPTUAL, DIAGNOSTIC, AND CROSS-MODAL EVIDENCE FOR MUSIC EVALUATION

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ABSTRACT

Evaluating generated music requires modeling perceptual quality, technical audio quality, and prompt alignment. Existing methods often overlook signal-level defects, produce compressed quality scores, and miss fine-grained musical attributes in text prompts. We propose *CloudEar*, a pairwise framework for evaluating musicality and prompt alignment. It comprises three components: *i*) a Perceptual Expert for song-quality assessment; *ii*) a Diagnostic Expert for learned and signal-level quality analysis; and *iii*) a Cross-modal Expert for prompt–song alignment based on musical attributes. Experiments show that *CloudEar* outperforms the compared baselines in pairwise preference accuracy and score correlation.

1. INTRODUCTION

Evaluating generated music requires assessing both audio quality and musical aesthetics. Signal-level defects such as clipping and abnormal loudness can degrade listening quality, while musical structure, expression, and performance require higher-level perceptual judgment [1]. Existing methods emphasize high-level musical semantics but often overlook technical quality and subtle perceptual differences. Regression models also exhibit central-tendency bias [2], resulting in compressed score differences.

In parallel, prompt alignment poses a separate challenge. General-purpose audio–text models capture broad semantic correspondence but often overlook fine-grained musical attributes such as instrumentation requirements. Consequently, high audio–text similarity does not always indicate whether a generated song follows the specific attributes requested by the prompt.

To address these limitations, we propose *CloudEar*, a pairwise framework for evaluating musicality and prompt alignment. Its Perceptual Expert combines ordinal regression, continuous regression, and pairwise ranking to model human-perceived song quality to mitigate prediction concentration. A Diagnostic Expert complements this representation with learned quality estimates and signal-

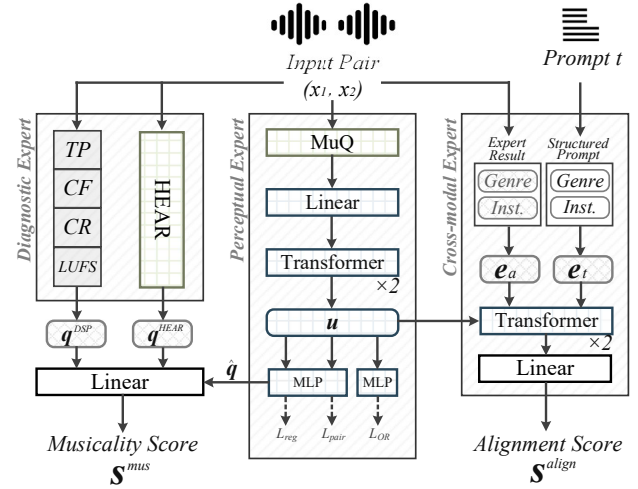


Figure 1. Overview of the proposed *CloudEar* framework.

level descriptors. For alignment, a Cross-modal Expert compares genre and instrumentation attributes extracted independently from the audio and prompt. Experiments on rating and pairwise-preference benchmarks show that *CloudEar* consistently improves both musicality and alignment evaluation over existing methods.

2. PROPOSED FRAMEWORK

As illustrated in Fig. 1, given a prompt t and two generated songs (x_1, x_2) , *CloudEar* predicts their pairwise preferences in musicality and prompt alignment.

2.1 Perceptual Expert

Following SongBench [3], we randomly sample a 30-second clip from each song x_i and extract MuQ features [4], which are processed by a Transformer adapter to obtain the perceptual representation u_i .

To alleviate the tendency of regression models to concentrate predictions around the mean [2], we jointly optimize ordinal regression, continuous regression, and pairwise ranking objectives. The ordinal loss L_{OR} [5] supervises ordered quality levels, while an MLP regression head predicts the quality score $\hat{q}_i$ using $L_{reg} = (\hat{q}_i - y_i)^2$, where y_i is the ground-truth score. For a training pair with $y_1 > y_2$, the pairwise loss encourages a higher score for



the preferred song:

$$L_{\text{pair}} = -\log \sigma(\hat{q}_1 - \hat{q}_2), \quad (1)$$

where σ denotes the sigmoid function. The overall training objective is

$$L_{\text{PE}} = 0.4L_{\text{OR}} + 0.4L_{\text{reg}} + 0.2L_{\text{pair}}. \quad (2)$$

2.2 Diagnostic Expert

The Diagnostic Expert complements high-level perceptual modeling with signal-level features and learned quality estimates. We extract true-peak level (TP), clipping ratio (CR), crest factor (CF), and integrated loudness (LUFS) to form the DSP feature vector $\mathbf{q}_i^{\text{DSP}}$, with loudness and true-peak measurements following standard practices [6, 7]. We also use HEAR [8], which combines MuQ and MusicFM representations, to obtain the quality score q_i^{HEAR} . The DSP feature vector, HEAR score, and perceptual score $\hat{q}_i$ are concatenated and fed into a linear layer to produce the musicality score s_i^{mus} .

The musicality fusion layer is trained using binary cross-entropy (BCE) between the pairwise preference label and the predicted probability $\sigma(s_1^{\text{mus}} - s_2^{\text{mus}})$. At inference time, *CloudEar* selects the song with the higher musicality score.

2.3 Cross-modal Expert

The Cross-modal Expert assesses genre and instrument alignment between the prompt and song. On the audio side, Genre Discogs400 [9] predicts 400 music styles, and MTG-Jamendo Instrument [10] predicts instrument tags. Their outputs are concatenated into $e_{a,i}$. On the prompt side, we use Qwen3-30B [11] to extract structured genre and instrument information from t . The extracted attributes are mapped to label vectors using the same vocabularies as the audio classifiers. These vectors are concatenated to form the prompt representation e_t .

A two-layer Transformer jointly models u_i , $e_{a,i}$, and e_t , followed by a linear layer that produces the alignment score s_i^{align} . The alignment branch is trained with BCE on pairwise preference probabilities obtained by applying a sigmoid to the score differences. For each pair conditioned on the same prompt, *CloudEar* selects the song with the higher alignment score.

3. EXPERIMENTS

3.1 Datasets & Metrics

We train the Perceptual Expert on AIME [12], MusicArena [13], and approximately 7,000 hours of in-house human-rated music, mainly English songs from five generation models. For the in-house dataset, we retain only songs rated by more than 200 listeners.

For pairwise training, we additionally use MusicEval [14] and the training split of CMI-Pref [15]. Their musicality annotations train the Perceptual Expert and musicality fusion layer, while

Table 1. Comparison and ablation results. Mus./Ali.: musicality/alignment. MuQ uses MuQ-Eval [17] for Mus. and MuQ-MuLan [4] audio-text similarity for Ali.

Method	CMI-Test		MA.	PAM-Music	
	Mus. P-ACC	Ali. P-ACC	P-ACC	SRCC	LRCC
CMI-RM [15]	.7833	.7389	.7194	.6626	.6379
HEAR [8]	.7882	.6970	.5769	.6180	.6230
SongBench [3]	.7340	.6823	.6746	.5437	.5123
MuQ [4, 17]	.5936	.6429	.6478	.3168	.3608
<i>CloudEar</i>	.8179	.7635	.7657	.7257	.7229
<i>PE + HEAR</i>	.7805	–	.7325	.6872	.7008
<i>PE + DSP</i>	.7620	–	.6845	.6351	.6626
<i>PE only</i>	.7592	.6743	.6736	.6056	.6515

their alignment annotations train the Cross-modal Expert. Musicality pairs use within-dataset rating gaps above 0.2 after $[0, 1]$ normalization while alignment pairs share a prompt.

Following prior work [15], evaluation uses PAM-Music [16], CMI-Test, and the MusicArena test set (MA.): 1,340 song pairs generated in Jul.–Dec. 2025, with no training-data overlap. We report SRCC and LRCC for scalar ratings and P-ACC for A/B preferences.

3.2 Experimental Results

Ablation Study. The full *CloudEar* outperforms all reduced configurations on musicality, confirming their complementary contributions. Both HEAR and DSP improve over *PE only*, demonstrating complementary information beyond the perceptual representation. The alignment gain over *PE only*, trained directly on alignment labels, supports the benefit of the Cross-modal Expert. The Perceptual Expert alone already outperforms SongBench on CMI-Test musicality and remains competitive on MusicArena-Test.

Comparison with Prior Work. For baseline evaluation, SongBench and HEAR use musicality scores for both tasks. *CloudEar* outperforms all compared baselines on the reported benchmarks. Its higher musicality and alignment P-ACC indicate better agreement with human judgments of quality and prompt alignment, respectively. Higher SRCC and LRCC on PAM-Music further demonstrate stronger rank and linear correlations with human quality ratings. Notably, SongBench and HEAR achieve above-chance alignment accuracy without prompts, suggesting a correlation between musicality and alignment preferences.

4. CONCLUSION

We presented *CloudEar*, a pairwise framework for evaluating the musicality and prompt alignment of generated songs. *CloudEar* integrates perceptual, diagnostic, and cross-modal experts to model song quality and fine-grained prompt alignment. Experiments on three human-annotated benchmarks demonstrate consistent improvements over existing methods. Future work will explain which musical factors lead to each preference.

5. AI USAGE STATEMENT

Qwen3-30B is used to extract structured genre and instrument attributes from input prompts, as described in Section 2.3. The authors take full responsibility for the content of this paper.

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